

Math 289, Fall 2026

Homework 3

Due Wednesday, September 23, 2026, 3:00pm, on Gradescope

Problem 1. Define the sequence $(a_n)_{n \geq 0}$ by $a_0 = 0$, $a_1 = 1$, $a_2 = 2$, $a_3 = 6$, and

$$a_{n+4} = 2a_{n+3} + a_{n+2} - 2a_{n+1} - a_n, \quad n \geq 0.$$

Prove that n divides a_n for all $n \geq 1$.**Problem 2.** Let d be a real number. For each integer $m \geq 0$, define a sequence $\{a_m(j)\}$, $j = 0, 1, 2, \dots$ by the condition

$$\begin{aligned} a_m(0) &= \frac{d}{2^m}, \\ a_m(j+1) &= (a_m(j))^2 + 2a_m(j), \quad j \geq 0. \end{aligned}$$

Evaluate $\lim_{n \rightarrow \infty} a_n(n)$.